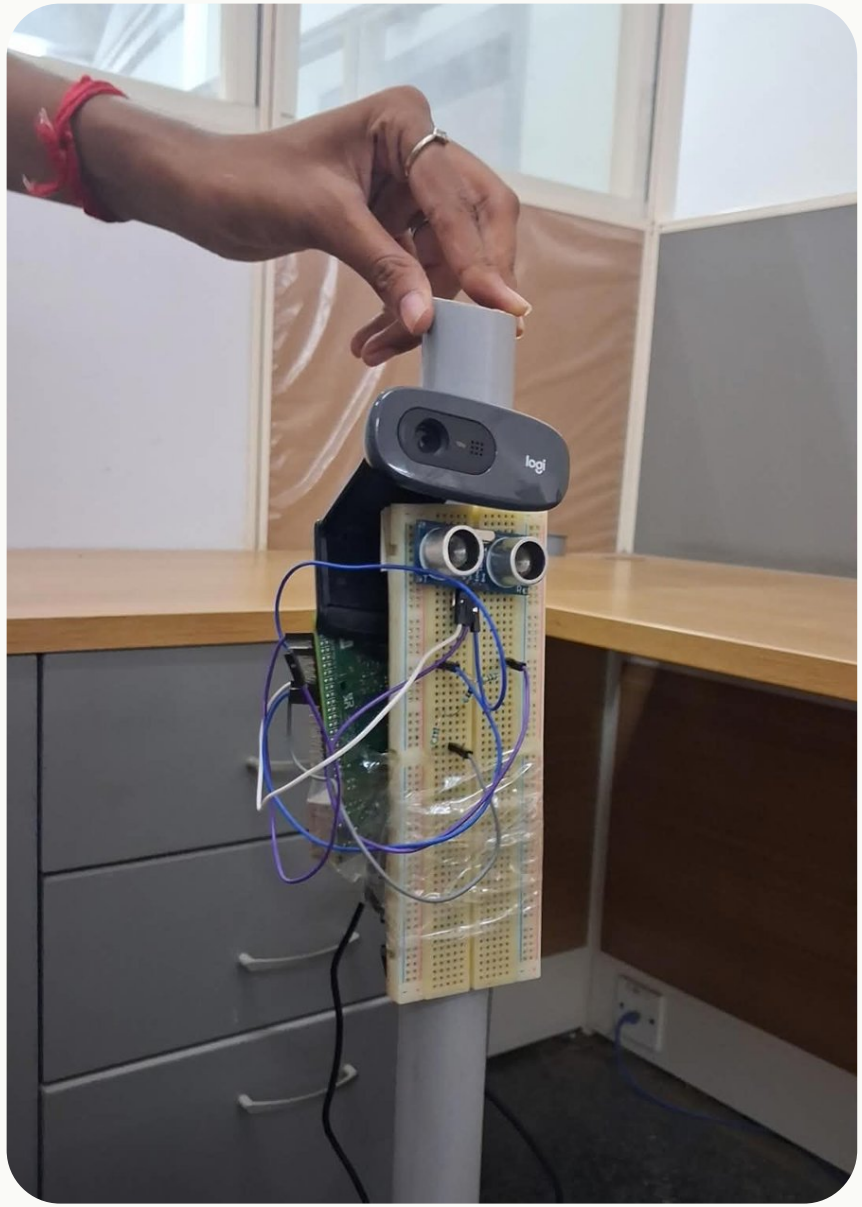
ARTIFICIAL & NEURAL NETWORKS · CAPSTONE

AURALIS

The smart cane that sees obstacles, speaks your language, and remembers where hazards are.

On-device AI | Hazard memory | 4 Indian languages | Hands-free



TEAM

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THE PROBLEM · AND WHAT EXISTS TODAY

A white cane feels only what it touches, and forgets it the moment it moves on.

2.2 B
people live with a vision impairment (WHO, 2019)

43 M
people are blind

Reach

Head-level boards, branches and fast bikes are outside the cane's sweep.

Memory

The same pothole on the daily route is found again, by touch, every time.

Language

Most aids speak only English; many users think in Hindi, Telugu or Tamil.

EXISTING SMART CANES vs AURALIS

Feature	WeWALK	Stanford	YOLO canes	AURALIS
Detects obstacles	Sensor	LiDAR	Camera	**Camera+sensor**
Names the object	No	Yes	Yes	**Yes**
Indian languages	No	No	No	**4, neural**
Remembers hazards	No	No	No	**Yes + decay**
Safest route	No	No	No	**Risk-aware A***
Approx. cost	High	$400	Varies	**Rs 8,300**

Gap: existing canes are reactive. None turn what they detect into a memory of the route and use it to warn early.

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OUR SOLUTION · TWO PILLARS

See it, say it, remember it.

PILLAR 1

AI hazard detection with voice

A YOLOv8 neural network on the Raspberry Pi names each obstacle, finds its side and distance, and speaks it in the user's language through the Pi or the phone's headphones.

“Chair detected ahead, 2 metres”

PILLAR 2

Location intelligence

Every confirmed hazard is stored with its GPS position on the cane. Repeated sightings strengthen it, time fades it, and the cane warns from memory before the hazard is in view.

“Careful. Remembered pothole, 25 metres ahead”

HOW IT WORKS

1 Sense
Webcam + ultrasonic + phone GPS

→

2 Detect
YOLOv8n on the Pi CPU

→

3 Measure
Size model + ultrasonic

→

4 Decide
Fusion MLP: 4 alert levels

→

5 Speak
Neural voice, 4 languages

→

6 Remember
Hazard map on the Pi

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THE PROTOTYPE

Built on a PVC pipe, in days.



Front: Logitech webcam, HC-SR04 below



Side: Raspberry Pi 4 on the shaft



Wiring: 1 kΩ / 2 kΩ divider on breadboard

WIRING (HC-SR04 → Pi 4)

Sensor pin	Pi pin
VCC	Pin 2 (5 V)
GND	Pin 6
TRIG	Pin 16 (GPIO23)
ECHO	1 kΩ → Pin 18 (GPIO24)
divider	2 kΩ GPIO24 → GND

MOUNTING

- Webcam at ≈ 0.9 m, facing forward, tilted slightly down
- Ultrasonic just below it, both 'eyes' clear
- Pi + power bank near the handle, ports open
- USB webcam, USB-C 5 V / 3 A power bank

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TECH STACK · HARDWARE AND SOFTWARE

Four parts on the cane. Three software layers.

HARDWARE

Raspberry Pi 4 — Rs 6,000
4-core ARM, 4 GB · runs the neural network, voice and memory

Logitech USB webcam — Rs 800
640×480 · forward vision at handle height

HC-SR04 ultrasonic — Rs 100
2–400 cm range · exact distance + hand gesture

Power bank — Rs 1,200
5 V / 3 A over USB-C

PVC pipe, breadboard, wires — Rs 200
The cane body

User's phone + headphones — —
GPS, app, voice in the ear

Total ≈ Rs 8,300

SOFTWARE

ON THE CANE (Pi)

Python 3 · ONNX Runtime · OpenCV · gpiozero · aiohttp WebSocket · edge-tts + mpg123

PHONE APP

HTML + JavaScript PWA · Leaflet + OpenStreetMap · GPS geolocation · Web Speech

TRAINING

Ultralytics YOLOv8 · NumPy MLP (backprop + Adam) · Google Colab GPU · Roboflow datasets

Pi → phone: obstacles over WebSocket ≈ 5×/s · phone → Pi: language, pause, saved places · memory on the Pi

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ARTIFICIAL NEURAL NETWORKS USED

Three networks, each with one job.

CNN · on the cane

YOLOv8n object detector

3.2 M parameters

- Convolutional backbone + detection head, pre-trained on COCO (80 classes).
- Exported to ONNX at 480 px; runs with ONNX Runtime on the Pi CPU, no PyTorch.
- Class-aware NMS (IoU 0.45, conf 0.40); object must appear in 2 of 3 frames.

MLP · trained by us

Sensor-fusion network

8 → 16 → 16 → 4

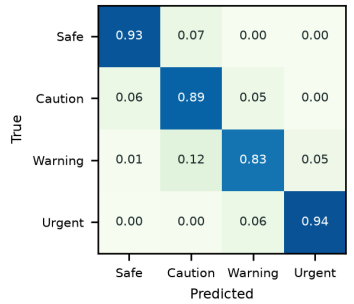
- Inputs: distance, drop, class severity, confidence, box area, offset, memory risk, approach speed.
- ReLU hidden layers, softmax output: safe / caution / warning / urgent.
- Backpropagation with Adam on 20,000 samples; safety override for drops.

Neural TTS

Natural voices

4 languages

- Neural voices: English (India), Hindi, Telugu, Tamil.
- Every sentence cached as MP3 on the Pi; ≈160 common ones pre-generated at start-up.
- Plays instantly and offline; espeak-ng only as a last fallback.



89.5%
alert-level accuracy (4,000 test samples)

0.94
urgent-alert recall

0 / 508
urgent cases predicted as safe

27 ms
YOLOv8 per frame, 4-core laptop CPU

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PILLAR 1 · AI HAZARD DETECTION

From a camera frame to a spoken warning.

640×480 webcam frame + ultrasonic 182 cm

↓ YOLOv8 on the Pi reads it

chair 0.81 | static obstacle | in path · ahead | 1.8 m

↓ neural voice, user's language

“Chair detected ahead, 2 metres”

“Stop. Chair very close” (under 0.8 m)

DISTANCE MODEL

d = H · f / h **f = (W/2) / tan(θ/2)**

H = typical object height, h = box height in pixels, θ = webcam field of view (calibrated once with a person at 2 m).
Straight-ahead objects take the ultrasonic distance instead.
Side = which third of the image the object is in.

HAZARD TAXONOMY

Hazard type	Examples	Response
Drop / pit	pothole, drain, stairs down	Always STOP
Static	chair, table, pole, bag	Warn + remember
Head height	branch, signboard	Warn + remember
Moving	person, bike, auto, dog	Warn only
Unknown	wall, box (ultrasonic)	Obstacle ahead

ANNOUNCED ONLY IF

- inside the 1 m walking corridor (middle third of the view)
- closer than 3 m
- seen in at least 2 of the last 3 frames (no one-frame false alarms)
- cane is active: hand at the sensor for 2 s pauses / resumes, no buttons

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PILLAR 2 · LOCATION INTELLIGENCE

Warn from memory, not from sight.

1 Remember

Each warned static hazard is saved with its GPS position, type and time, on the Pi (memory.json).

2 Weigh

Seen again → stronger. Not seen for a week → half as strong. The user can clear it.

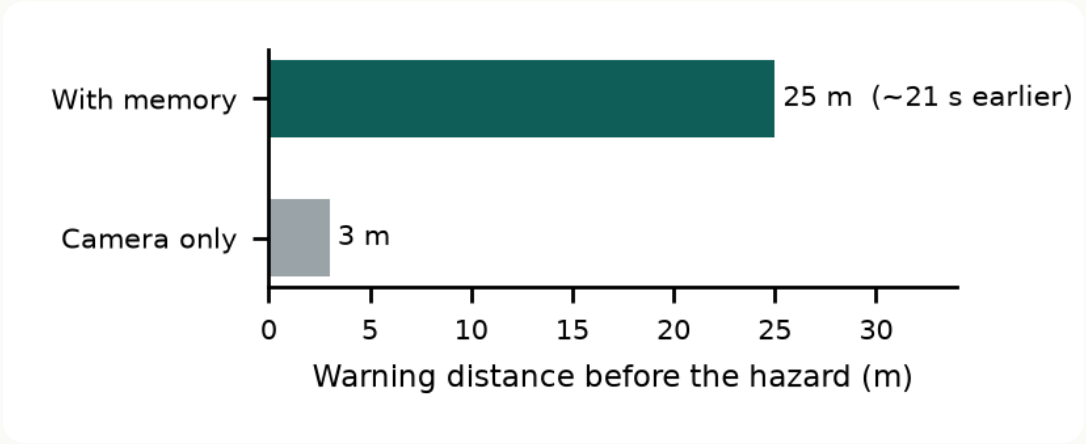
3 Learn the paths

GPS breadcrumbs every 5 m build a walking graph from the user's own routes.

4 Route safely

A* finds the route with cost = length × (1 + 4 · risk) to a saved place.

```
risk = severity × (1 − e^(−n/2)) × 0.5^(age/7d) × conf
```



8.3×
earlier warning (25 m vs 3 m)

21 s
extra reaction time at 1.2 m/s

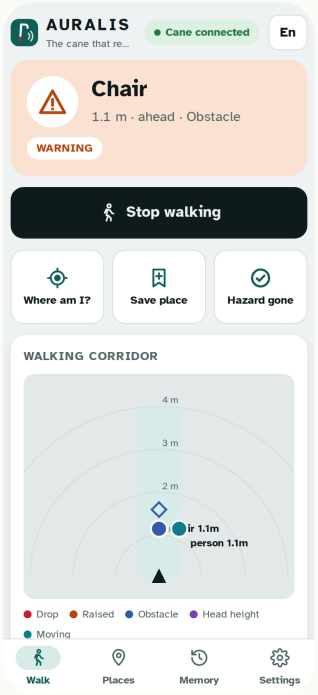
−60%
route risk for +12% length

Route test: shortest 195 m at 10% risk vs safest 218 m at 4% risk. Measured in the app's route simulation; outdoors the phone's GPS (±5–10 m) is used.

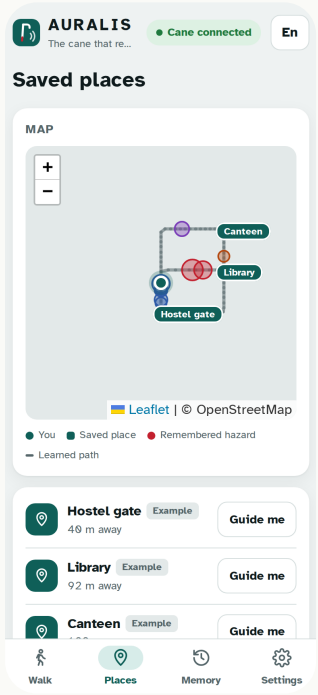
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THE PHONE APP · AND RESULTS

Headphones on, nothing to press.



Walk: live status + radar



Places: map + memory



Memory, in Hindi

English **Hindi** **Telugu** **Tamil**

Every screen and every voice alert in 4 languages; voice through phone headphones.

RESULTS

Test	Result
Detector vs reference YOLOv8	Identical classes and boxes
Detector speed, laptop CPU	13 ms (320 px) · 27 ms (480 px)
Fusion MLP, 4,000 test rows	Accuracy 89.5% · urgent recall 0.94
Early warning from memory	25 m ahead vs ≈ 3 m camera-only
Safest vs shortest route	4% vs 10% risk, +23 m
Languages	EN / HI / TE / TA, switch live
Hands-free pause	2 s hand hold, no false toggles

Distance accuracy and Pi 4 speed: measured on the cane with pi/measure.py and reported in the IEEE paper.

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FUTURE IMPLEMENTATIONS · AND THANK YOU

From a cane to a full wearable.

SMART CANE 2.0

- Custom YOLO: potholes, open drains, stairs, curbs
- Downward sensor for drops and steps
- Haptic handle: left / right / stop
- Shared hazard map across many canes

SMART GLASSES

- Camera at eye level for head-height hazards
- Bone-conduction audio, ears stay free
- Read signs, bus numbers, currency
- Scene description on demand

LEARNING

- Retrain fusion on real labelled walks
- Learned risk model instead of a formula
- INT8 model for faster frames
- Field trials with blind users

READ MORE

IEEE paper	AURALIS_IEEE_Paper.pdf (GitHub)
Code	github.com/Lakshya0104/Auralis-
App demo	claude.ai/artifact/8sLp1jaMJCcjauKUYCU8F8

Thank you

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